Bringing ML to production with Cloud

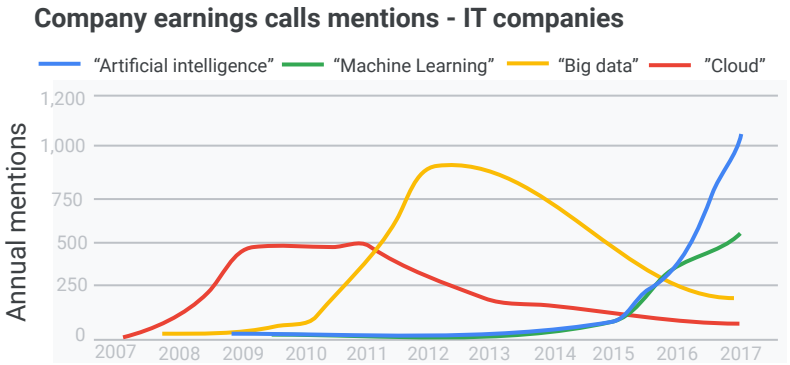
Google Cloud AI Platform

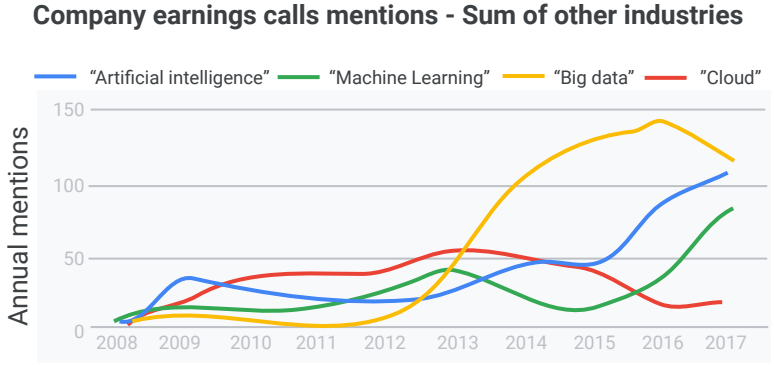
Javier Cañadillas - Enterprise Architect - **javiercm@google.com**

Google Cloud



Every company is talking about AI today





Source: AI Index Report 2018

Google Cloud

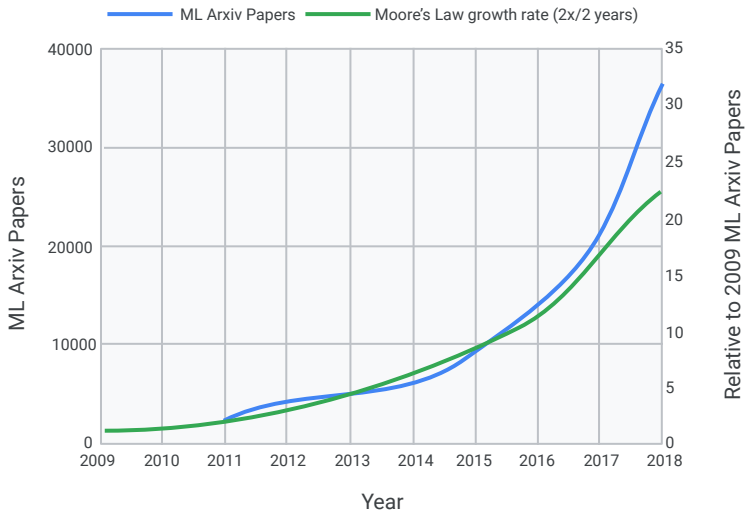
When most companies talk about AI in their transformation journey, you hear terms like...



Google Cloud

We are at the peak of AI's hype

Research



Reality

12%

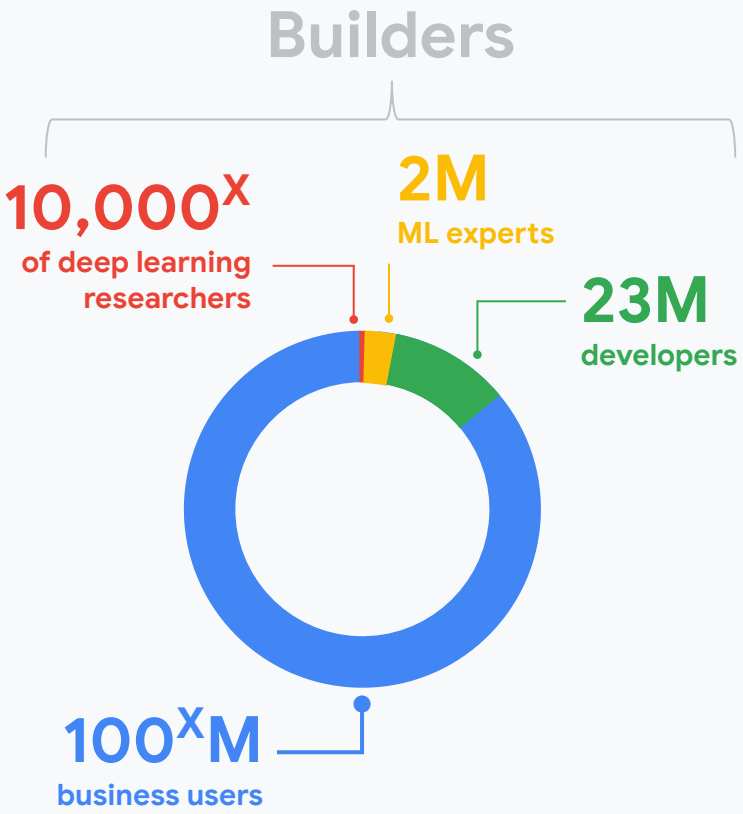
Of enterprises have deployed AI in production[1]

[1] McKinsey Global Institute, *Artificial Intelligence: The Next Digital Frontier*

Google Cloud

2019.

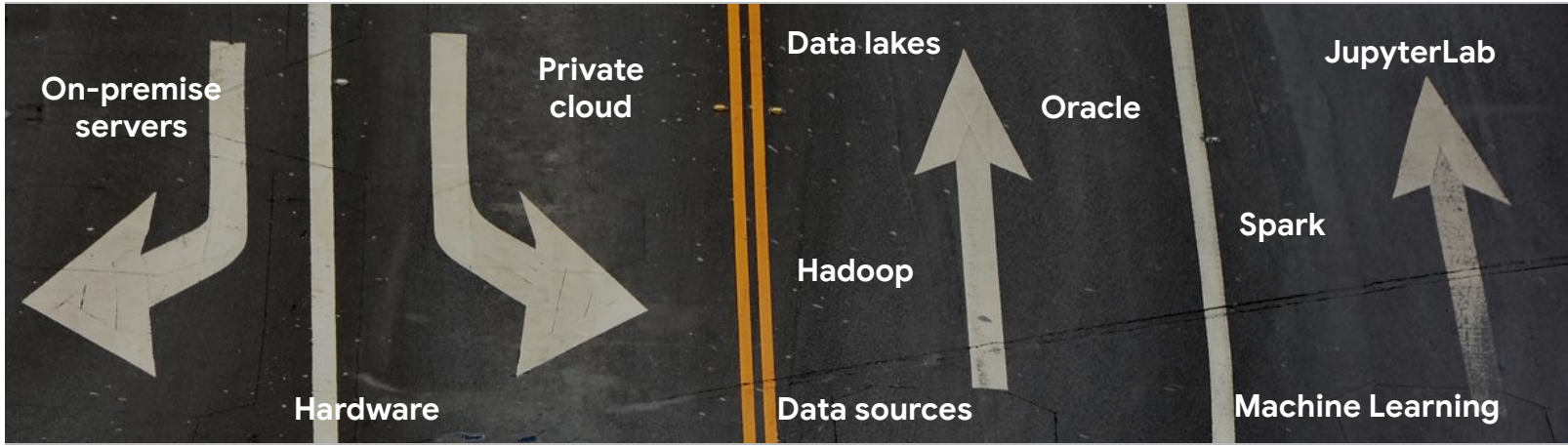
Empower every enterprise to **transform** their business with AI



Google Cloud

Enterprise challenges

How do I get the most out of the data, with my existing infrastructure and people



Google Cloud

Let Cloud platforms help



Modernize your IT, either on Cloud, on-premises, or both

Use packaged migration offerings to build your data warehouse on the Cloud (i.e.: BigQuery). Run Kubeflow on premise to manage all machine learning workloads and scale to the Cloud seamlessly as needed.



Foster team collaboration to solve for impact

Help your ML developers and data scientists encapsulate their work for reuse with Kubeflow Pipelines, and share it with other application developers, data engineers, and business analysts who can discover and deploy it easily using AI/ML catalogs like AI Hub.



Embed AI into applications at production scale

Move your models from experimentation to production using the tools provided by platforms linke AI Platform to abstract underlying infrastructure and operational dependencies, and explain outcomes driven by AI to business users.

Google Cloud

Data warehousing for AI-driven business



Data warehouses

From 1st-gen EDWs, increased data collection and analysis has helped build more data-driven businesses.

BI foundations

Data warehousing formed the foundation of reporting and business intelligence.

Cloud data warehousing

Fully managed, analytic systems like BigQuery represent a fundamentally different approach to cloud data warehousing.

AI foundations

Cloud Analytic engines to work as the foundation for organizations that will leverage machine intelligence in their businesses.

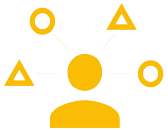
Google Cloud

Cloud Analytics systems form the AI foundation



Automate data delivery

Automated data transfer to extract data from your systems & shared data with federated querying across any Google service



Democratize data insights

Break data silos, power apps, add read-only data sets & make query results accessible to anyone



Build the foundation for AI

Enterprise Data Warehouse stores the most valuable data for your company & brings AI capabilities without replicating data into storage



Tee up real-time insights

Analyze real-time business events by automatic data ingestion, which is immediately available to query in your data warehouse

Google Cloud

AI is a team sport. How can we connect users to produce work that is 10x more impactful?













Google Cloud

The flywheel of AI adoption with ML Catalogs

Search & Discover

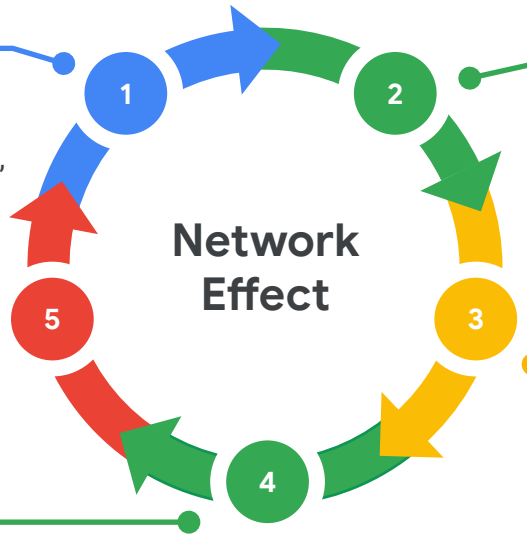
Find best-of-breed pipelines on the hub which leverage Cloud AI Platform solutions (AutoML, TPU, CMLE, etc.).

Publish

Upload & share pipelines with your organization or publicly.

Run in Production

Deploy customized pipelines in production on Cloud or on-premise.



Deploy

Easily deploy ML pipelines onto the Cloud or on-premise.

Customize

Experiment and append out-of-the-box pipelines to custom use cases via pipelines UI.

Google Cloud

AI/ML will be synonymous with software



Decision trees

Engineers applied decision trees to drive machine outcomes.

Modeling and simulation

Faster computers and software paved the way to apply statistics to drive superior outcomes.

Deep Learning

Deep Learning ushered the possibility to solve previously unsolvable problems.

Embedded Intelligence

Businesses view AI as an integral part of product development and operational efficiency.

AI/ML Platforms helps you...



Build once to deploy anywhere

Support Kubeflow, Google's open-source technology to build portable AI applications that can run on-premises or on Google Cloud without significant code changes



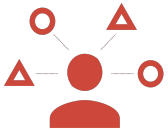
Productionize AI applications fast

Help move models into production in a serverless way, thereby allowing your team focus on development, not on operations



Explain and optimize AI easily

Provide services to validate training data, understand model outcomes, and improve model accuracy through continuous learning



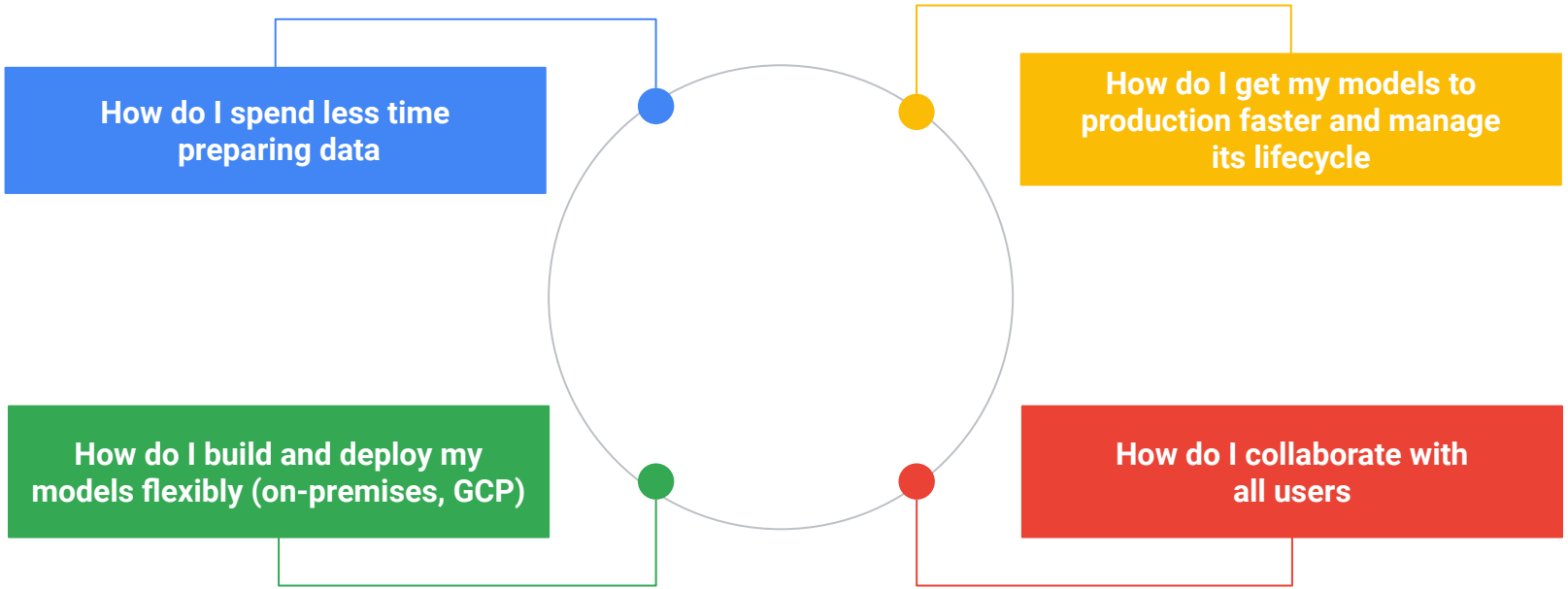
Collaborate effectively

Access the best of AI content built with users across the organization, as well as AI/ML Catalogs

Google Cloud

Technical challenges & Cloud Solutions

How can I make faster impact on business



Google Cloud

Production ML/AI
go beyond Data Science



Pick your flavor

ML APIS

Ready to go



Cloud Vision



Text to Speech



Speech to Text



Cloud Translation



Cloud Natural Language



Cloud Video Intelligence

Pick your flavor

AutoML

Your data, our models



- Vision
- Video
- Natural Language
- Translation
- Tables

ML APIS

Ready to go

Cloud Vision

Text to Speech

Speech to Text

Cloud Translation

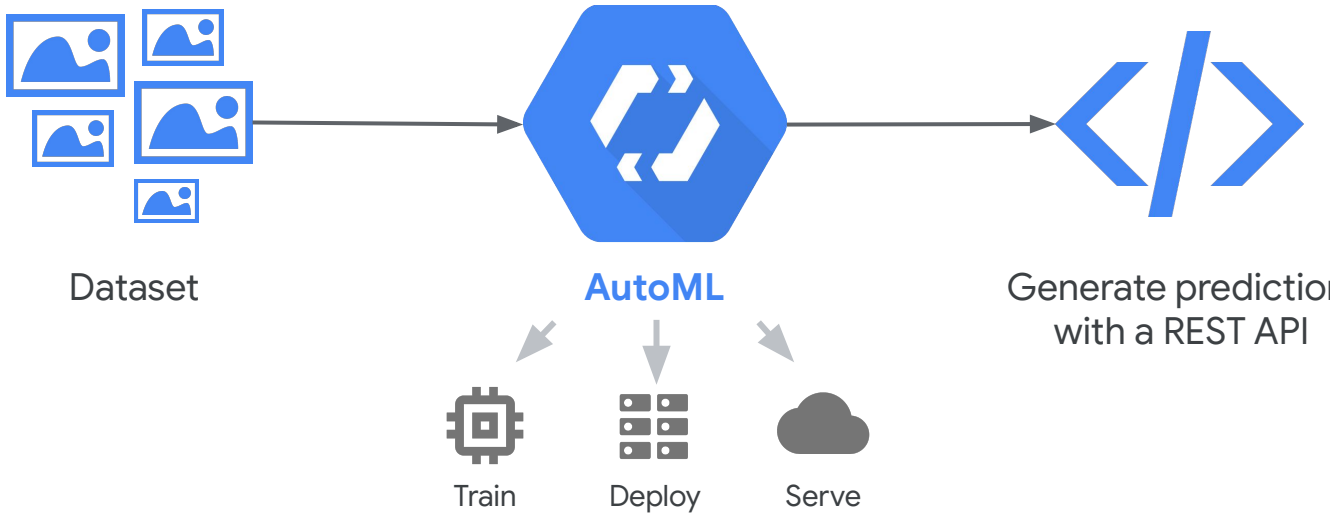
Cloud Natural Language

Cloud Video Intelligence

Google Cloud

AutoML

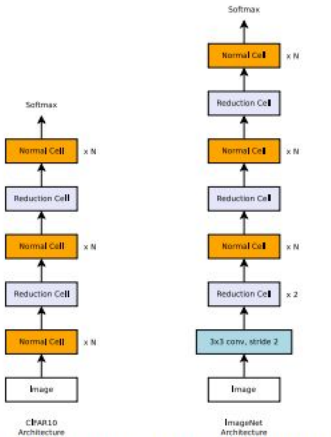
ML that creates ML for your problem



Google Cloud

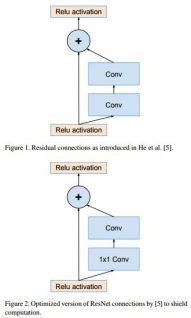
AutoML

Learn2Learn



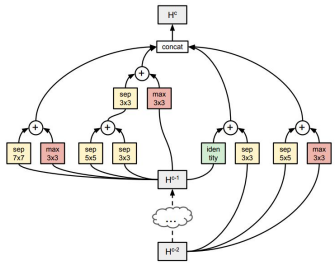
Learning Transferable Architectures for Scalable Image Recognition, Barret Zoph, Vijay Vasudevan, Jonathon Shlens, and Quoc V. Le. Arxiv, 2017.

Transfer Learning

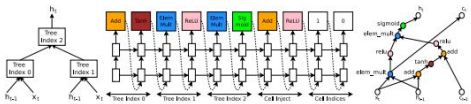


Inception-v4, Inception-ResNet and the Impact of Residual Connections on Learning Christian Szegedy, Sergey Ioffe, Vincent Vanhoucke, and Alex Alemi. AAAI, 2017.

Hyperparameter Tuning



Progressive Neural Architecture Search Chenxi Liu, Barret Zoph, Jonathon Shlens, Wei Hua, Li-Jia Li, Li Fei-Fei, Alan Yuille, Jonathan Huang, Kevin Murphy, Arxiv, 2017



Barret Zoph, Quoc V. Le. ICLR 2017.

[Large-Scale Evolution of Image Classifiers]

Esteban Real, Sherry Moore, Andrew Selle, Saurabh Saxena, Yutaka Leon Suematsu, Quoc Le, Alex Kurakin. International Conference on Machine Learning, 2017.

[Bayesian Optimization for a Better Dessert]

Benjamin Solnik, Daniel Golovin, Greg Kochanski, John Elliot Karr

Pick your flavor

ML Frameworks

Total Control for Power Users





AI Platform



Spark ML



Cloud Dataproc
(managed Hadoop)

AutoML

Your data, our models

Vision

Video

Natural Language

Translation

Tables

ML APIS

Ready to go

Cloud Vision

Text to Speech

Speech to Text

Cloud Translation

Cloud Natural Language

Cloud Video Intelligence

Google Cloud

But let's focus on Data Science

Data Science Problems Today

1. Infrastructure

Managing infra: Managing infrastructure can be challenging for many (especially scaling and distributing)

Hybrid: Brittle, opinionated infrastructure that is hard to productionize and breaks between cloud and on-prem

2. Collaboration & Talent

Talent: Machine learning expertise is scarce / 10x your ML experts

Speed of innovation: Experimentation and reuse is limited by tools

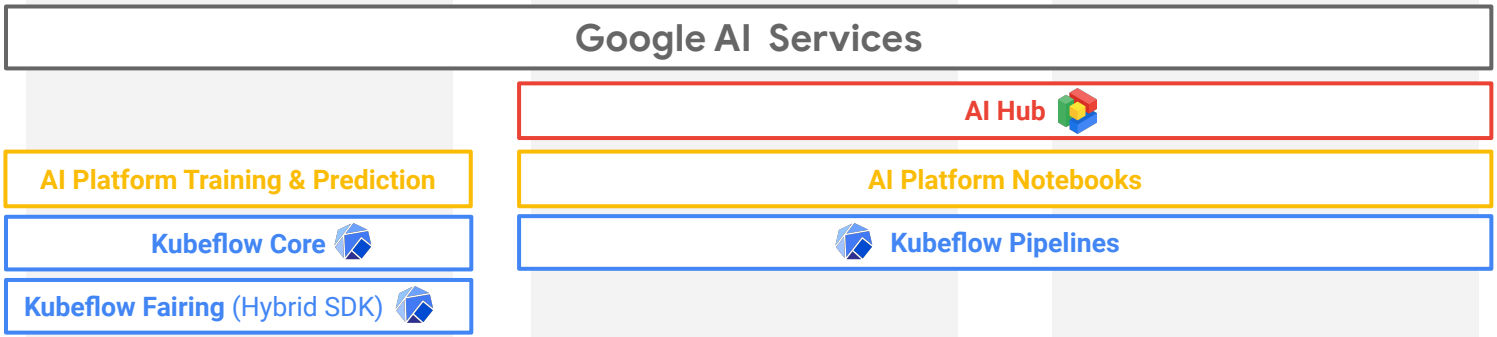
Internal COE: Difficult to find, leverage existing solutions within your organization

3. Getting started

Setting up: Setting up and maintaining a stack optimized for Deep Learning is hard and time consuming

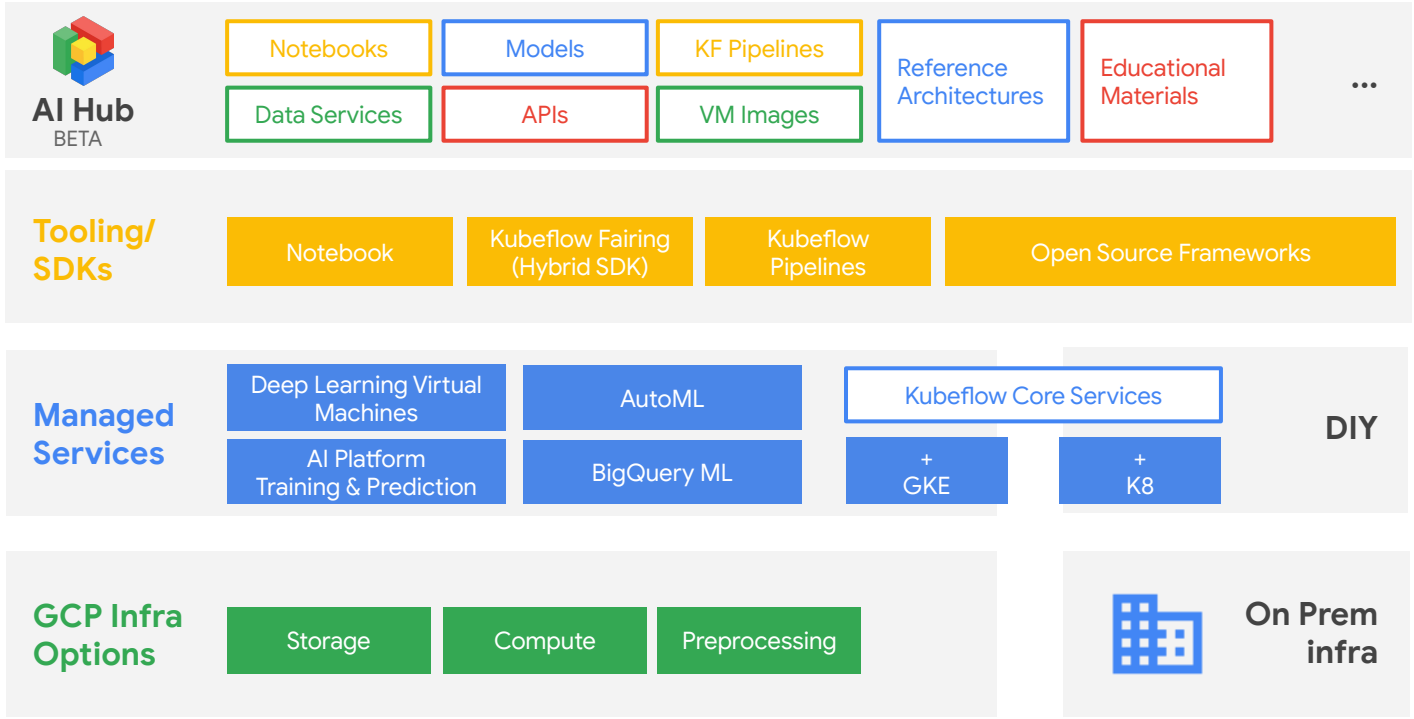
Samples: Finding relevant samples and components

Data: Access to high quality, ready to use data for ML



Google Cloud

Example: Google Cloud Platform ML Stack



Google Cloud

Prepare your data

Ingest

Ingest and distribute data reliably



Cloud Pub/Sub



Apache Kafka

Transform

Fast, correct computations quickly and simply



Apache Beam

Cloud Dataflow



Apache Spark

Label

High quality training data for machine learning



AI Platform
Data Labeling

Google Cloud

Cloud Dataflow

The fully-managed data processing service that simplifies development and management of stream and batch pipelines

Accelerate development for streaming & batch

Fast, simplified data pipeline development via expressive Java, Python & Go APIs in the Apache Beam SDK

Simplified management and operations

Remove operational overhead by letting Cloud Dataflow auto-manage performance, scaling, availability, security and compliance.

Build on a foundation for machine learning

Add TensorFlow-based Cloud Machine Learning models and APIs to your data processing pipelines for real-time predictions

Google Cloud

Data Labeling

Get high quality training data

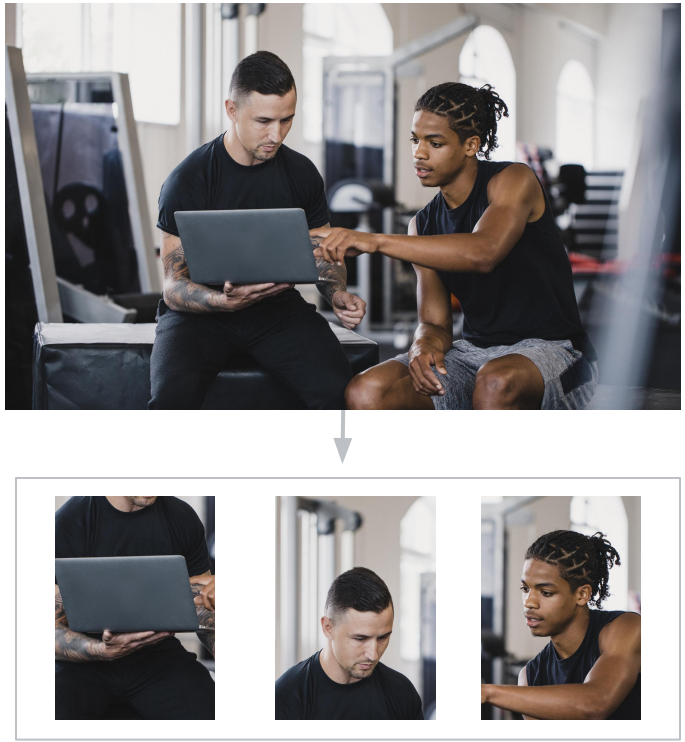
Supports the most popular use cases of image, video, audio, and text annotation with quality assurance.

Choose with confidence and ease

Allows you to create a plan for our qualified, trusted human labelers to annotate your data with instructions and examples.

Work seamlessly

Import your labeled data to AutoML to continue with your machine learning development.



Google Cloud

Start Building Quickly

Configured environments

Get a head start with fully configured VMs for popular data science and deep learning frameworks



Deep Learning VMs

Built-in models

Use the built-in models inside AI Platform to develop with and deploy



AI/ML Cloud Platforms

ML Pipelines and more

Build onto end-to-end ML pipelines, notebooks, and related content from the AI Hub



AI/ML Catalogs

Google Cloud

Deep learning turn key VM images

Easier and faster ML

- **Fast prototyping**
  Prototype your ML project quickly with pre-configured VMs for deep learning.
- **CPU, GPU and TPU support**
  Choose to add the latest Cloud TPU or GPUs to your instance in a single click and accelerate your model training jobs.
- **Performance optimized for Google Cloud**
  We tune the libraries and config to get the optimal performance on our infrastructure, so you don't need to worry about it.
- **Flexibility**
  Choose between different ML frameworks like TensorFlow, PyTorch, RAPIDS, and scikit-learn or install your own on our common base image.



Google Cloud

Fully managed notebook with configured environments

Pre-installed GCP client libraries

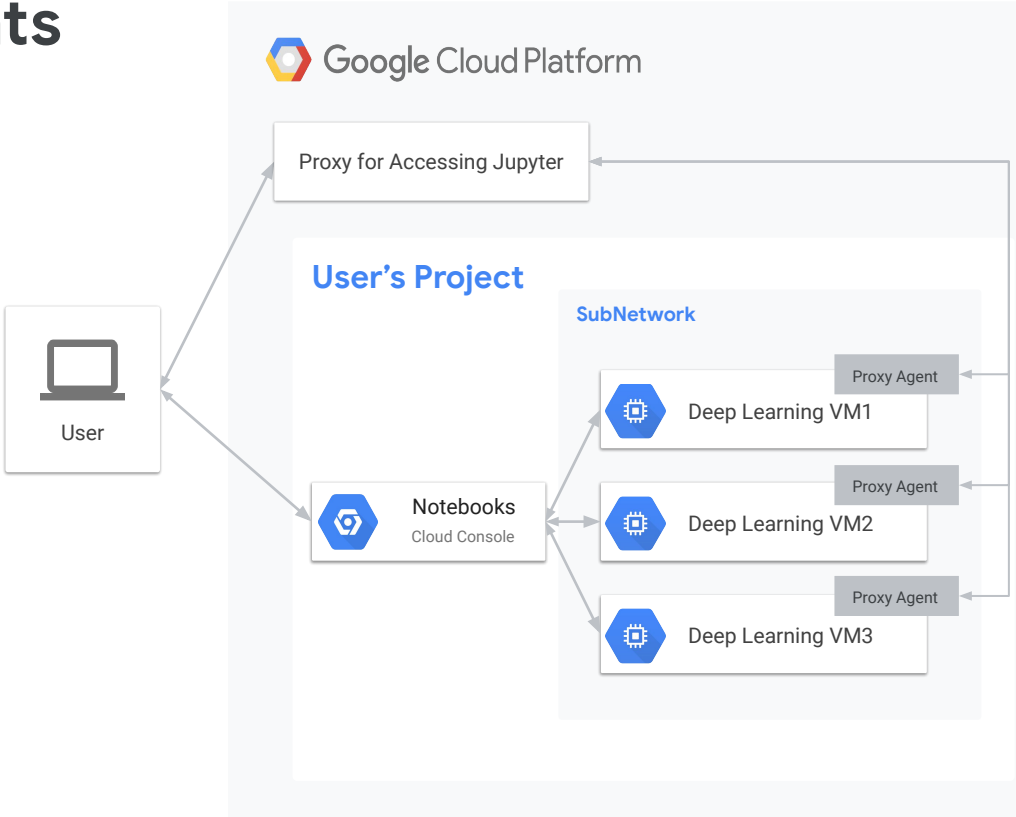
Pre-configured environments for:

NumPy SciPy scikit learn

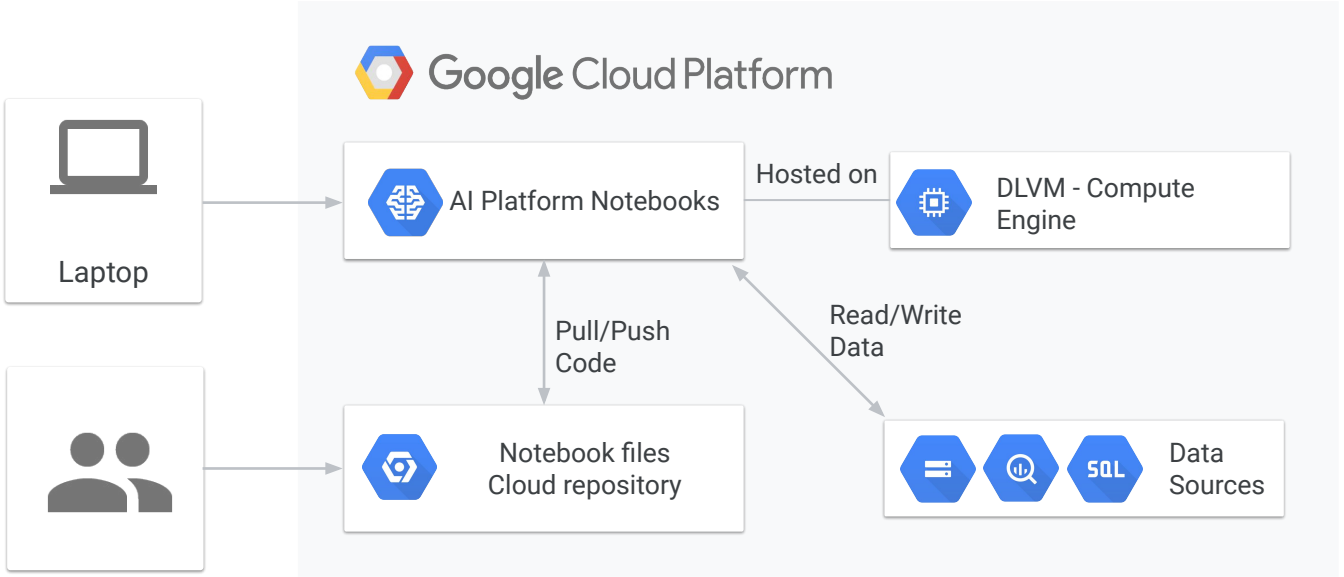
matplotlib TensorFlow Keras

dmlc XGBoost PyTorch

Google Cloud

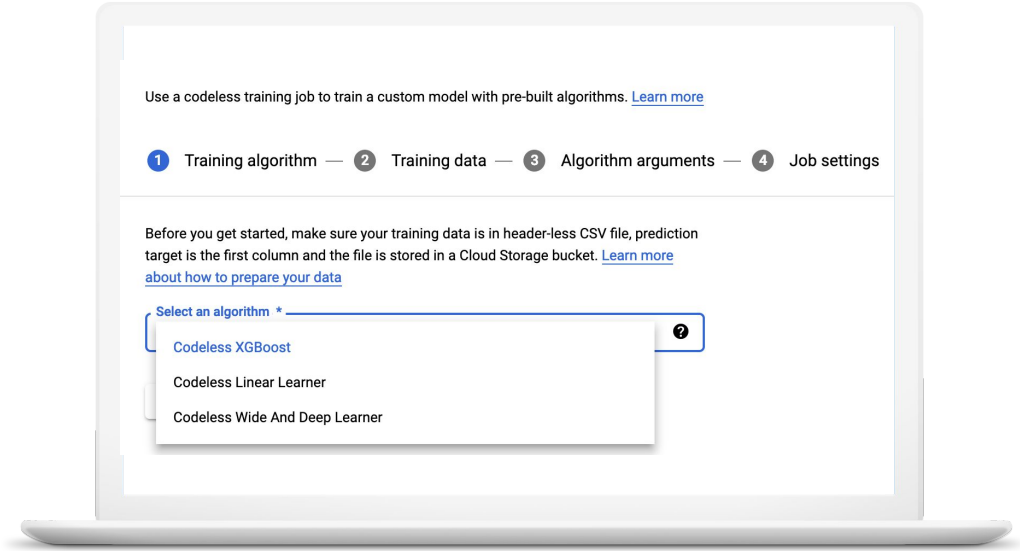


How do managed Notebooks work?



Built-in models in AI/ML Platforms

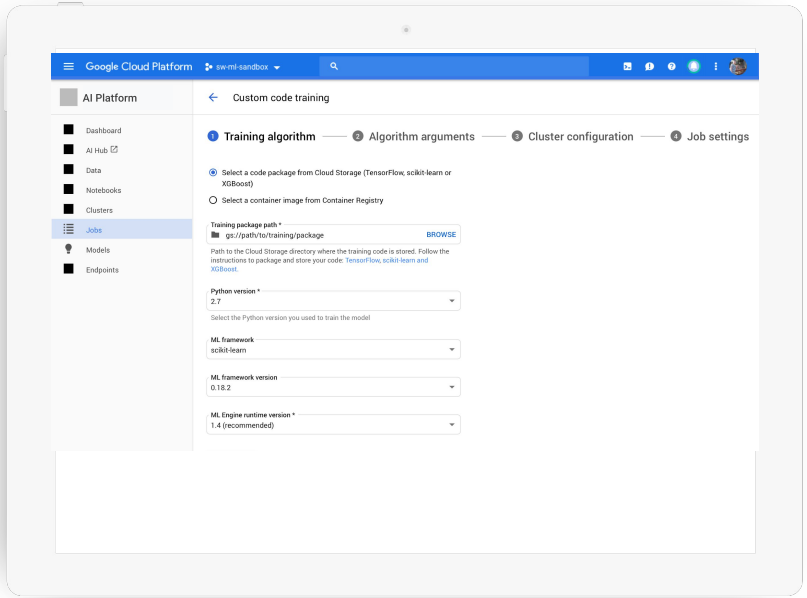
- Integrated Hyperparameter Tuning
- Support for popular algorithms
- Discoverable via AI/ML Catalogs
- Easy to add new algorithms
  Algorithm as a container



Google Cloud

Serverless training using AI/ML Platforms

- Train models without finagling with infrastructure
- Supports all popular data science and machine learning frameworks. You can even run your own Docker container
- Leverage distributed training on the latest GPUs and TPUs to finish jobs faster
- Improve your model quality with the state-of-the-art automated hyperparameter tuning



Google Cloud

On-premises training using Kubeflow

Infrastructure abstraction

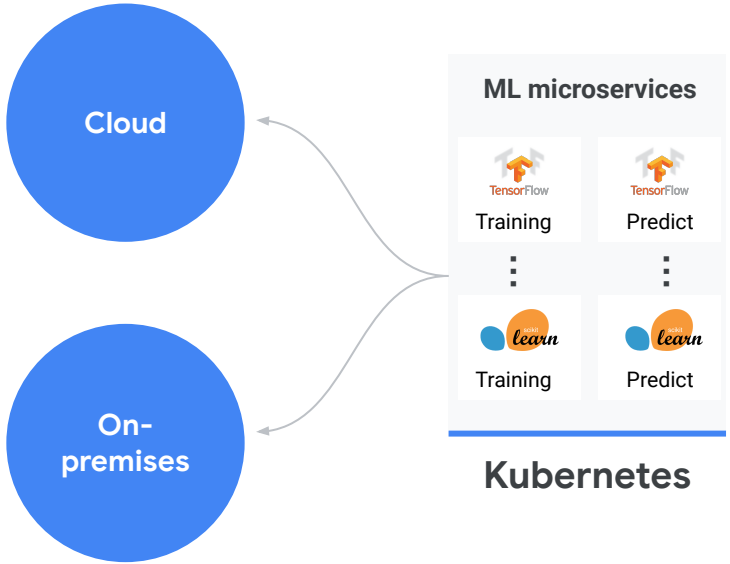
Kubernetes manages all underlying dependencies, and resources

Swappable & scalable

Library of ML microservices to deploy training and prediction jobs

Run where you want

- Cloud
- On-premises



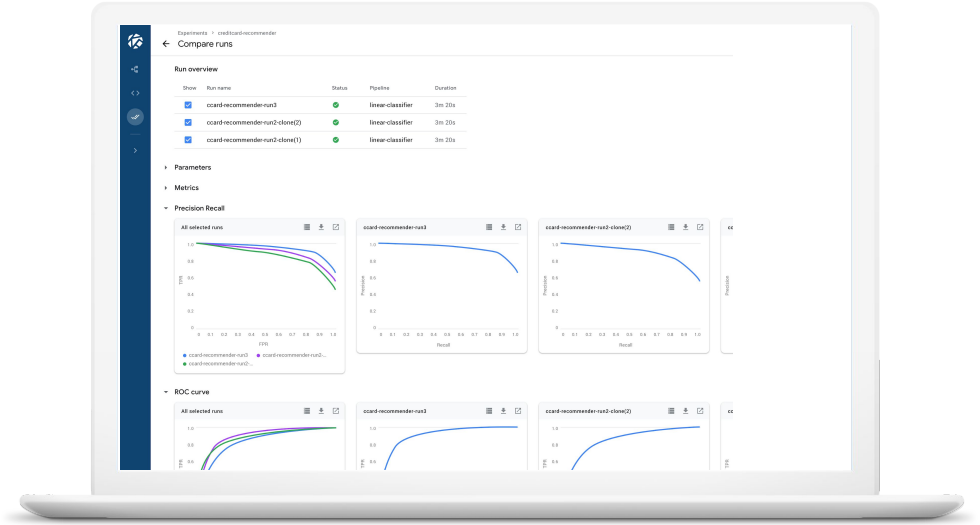
Google Cloud

Manage and analyze models

Integrated TFX tools to analyze models for bias and drift

Visual UIs to compare different versions of a model

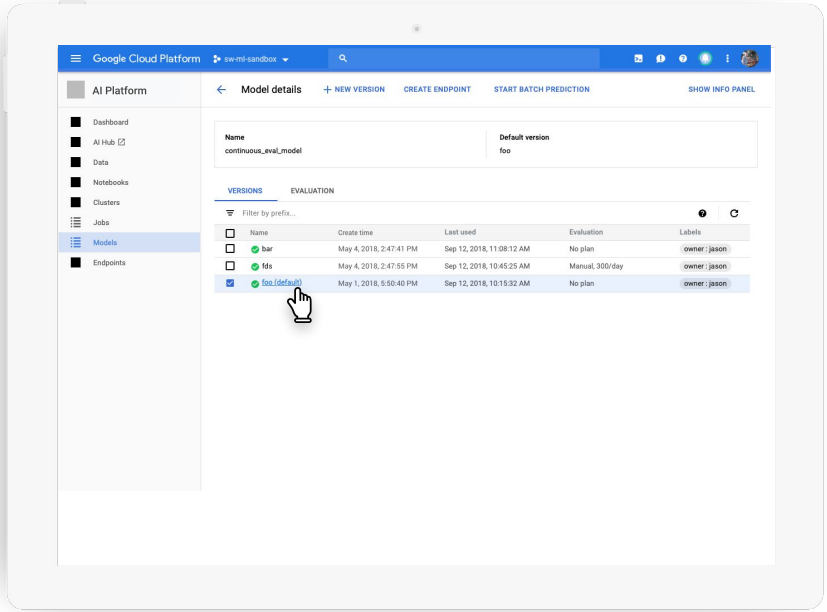
- Plot metrics side by side or in the same chart
- Load TF metrics in same TensorBoard instance
- View diff configs, parameters and even code changes from the console



Google Cloud

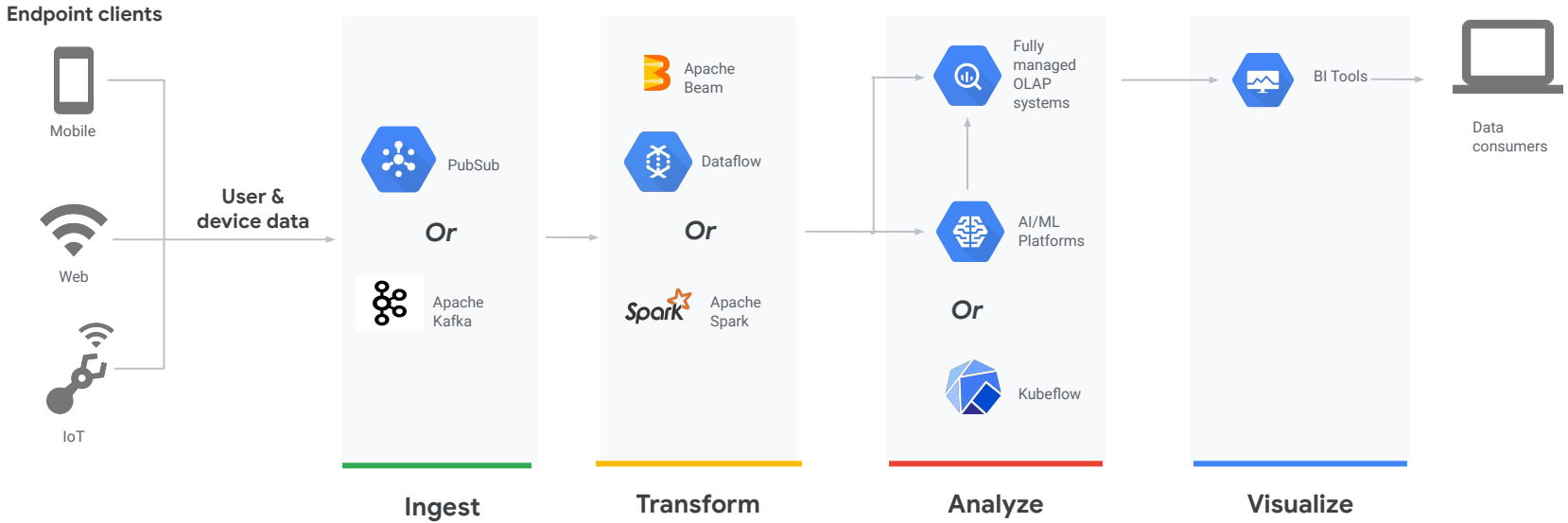
Deploying models with ease

- Set up **online** endpoints for low-latency predictions, or get predictions on massive **batches** of data
- Deploy models trained on premises or on Cloud
- Scale automatically based on your traffic
- Use GPUs for faster predictions



Google Cloud

Accelerate your ML development with our unified, open and fully-managed architecture

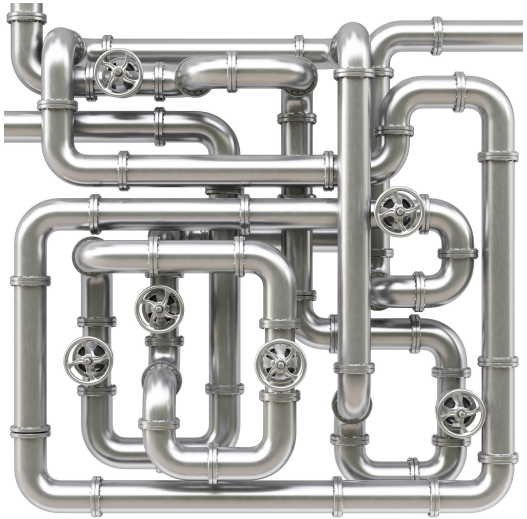


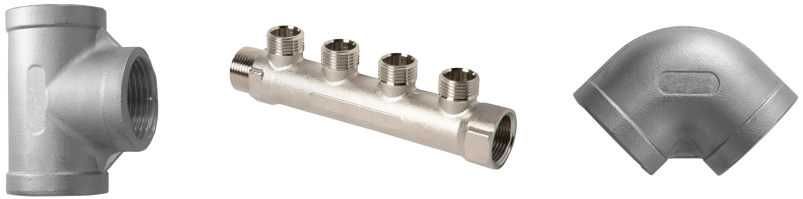
Google Cloud

Publish your machine learning workflow in an orchestrated pipeline

Build custom AI applications by easily “stitching” and connecting various components like LEGO.

Publish end-to-end pipelines that can be reused by other developers without worrying about the underlying details





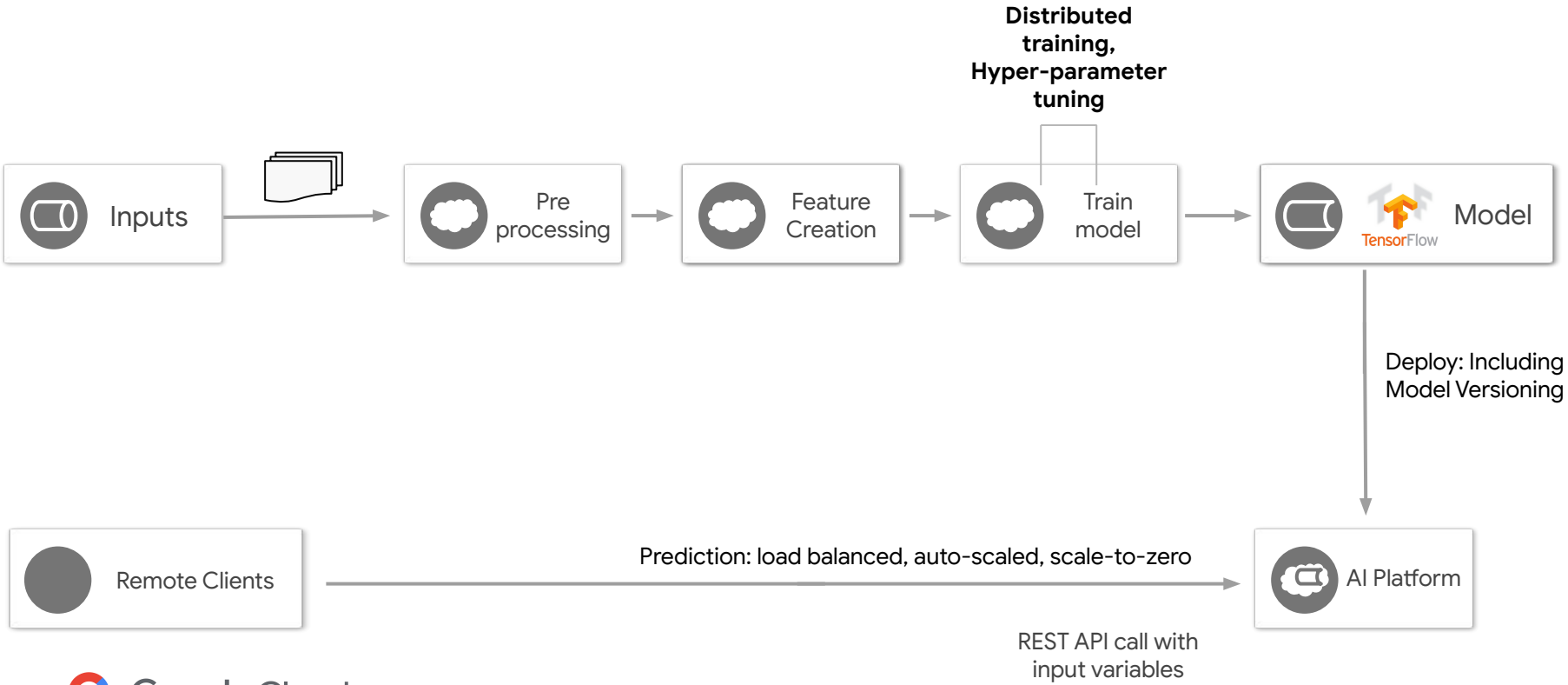
Google Cloud

Taking ML models into production

"Naive" view of a ML production pipeline

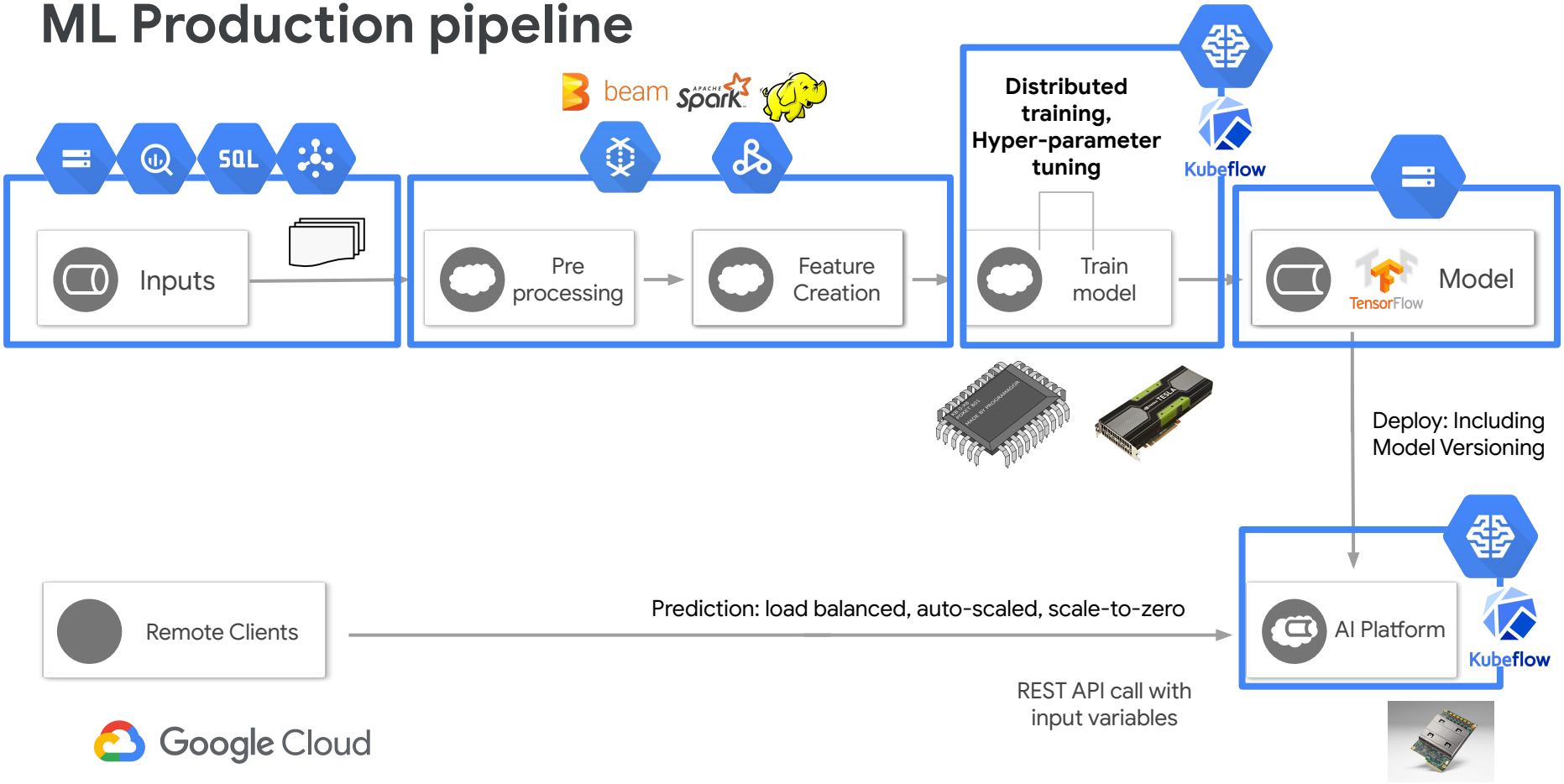


ML Production pipeline

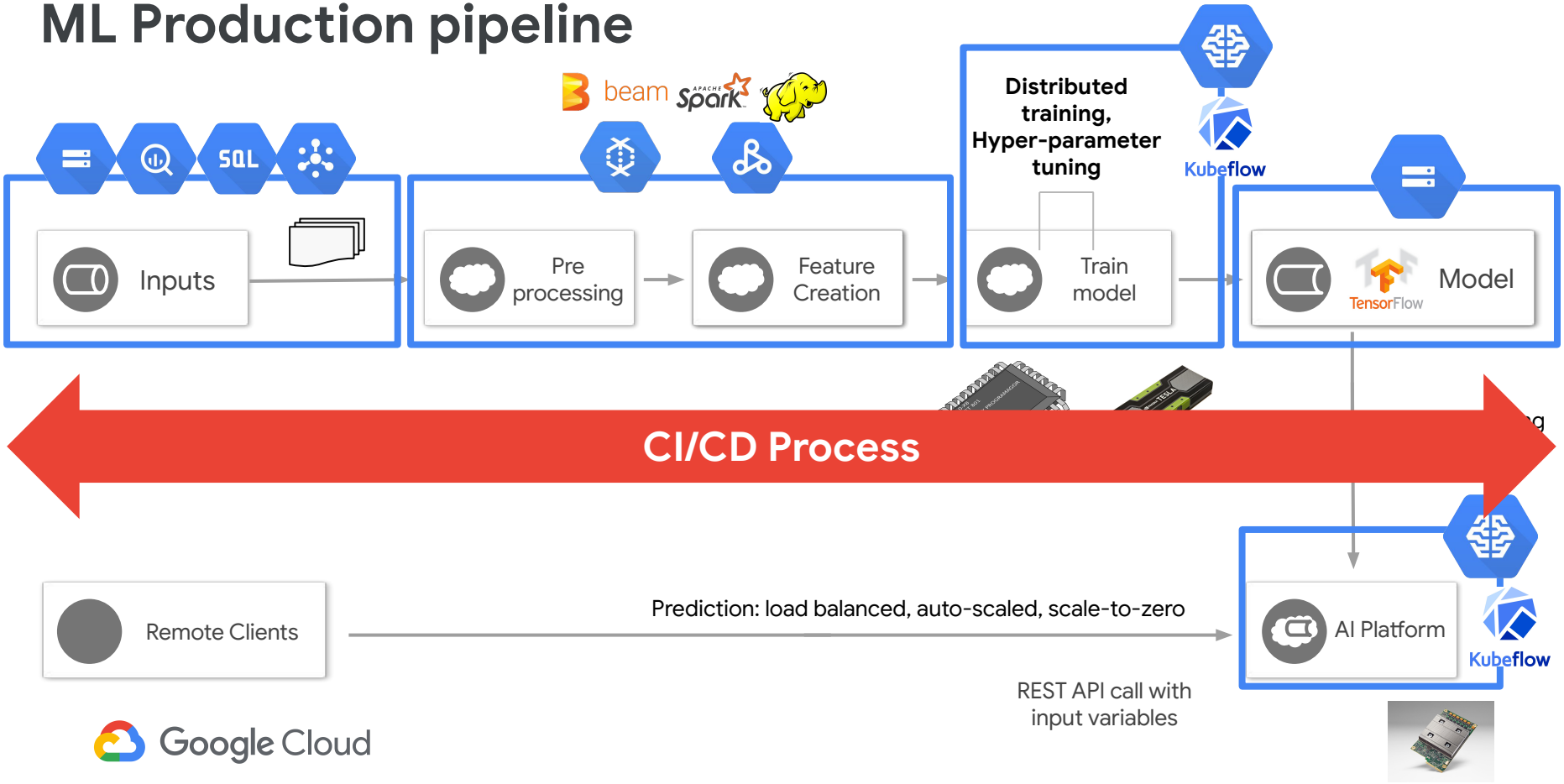


Google Cloud

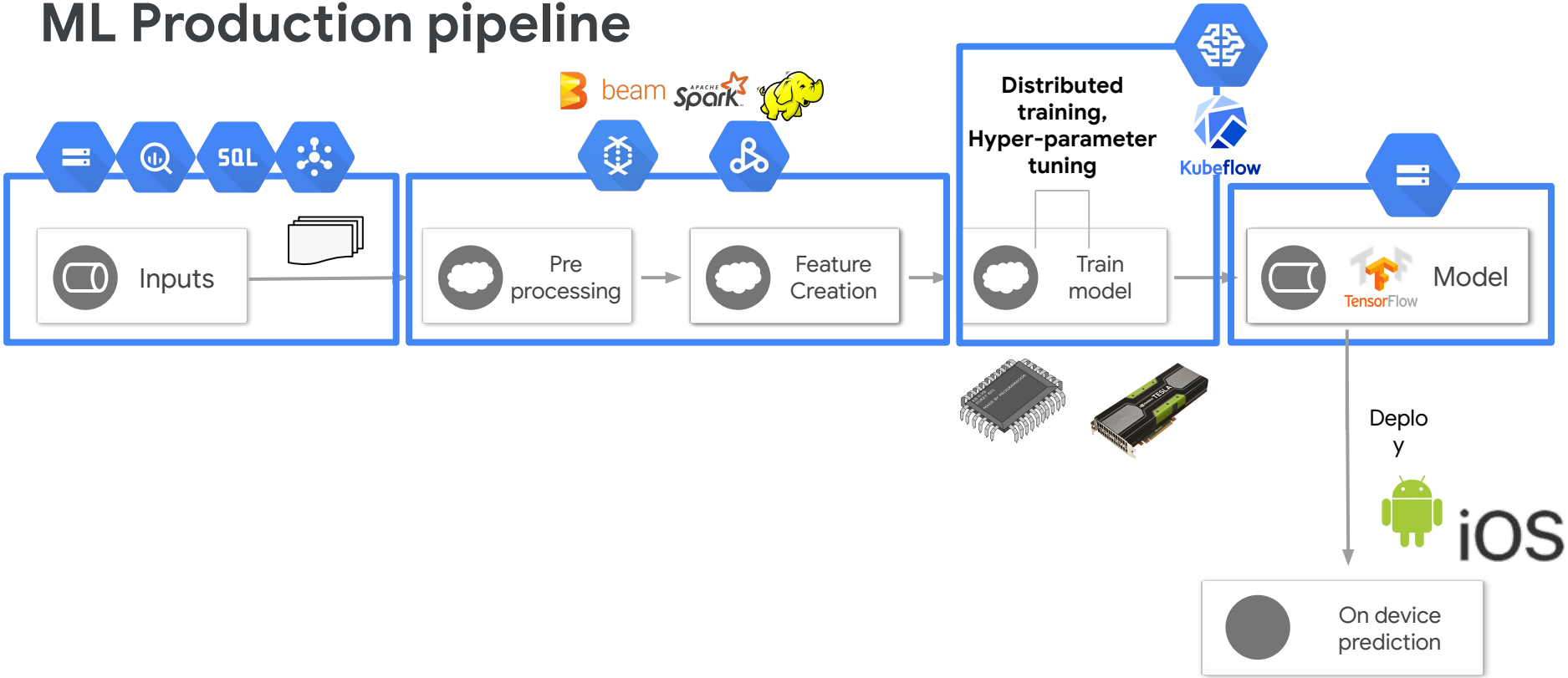
ML Production pipeline



ML Production pipeline



ML Production pipeline



Google Cloud



¡Gracias!

Google Cloud